

Question				Marking details				Marks available																							
								AO1	AO2	AO3	Total	Maths	Prac																		
3	(a)	(i)		Red variegated RRGA RG, RA				Red/white variegated RWGA (1) RG, RA, WG, WA (1)					1 1		2		2														
		(ii)		<table><tr><td>RRGG</td><td>RWGG</td><td>RRGA</td><td>RWGA</td><td>RRAA</td><td>RWAA</td></tr><tr><td>Red green</td><td>Red/white green</td><td>Red variegated</td><td>Red/white variegated</td><td>Red white</td><td>Red/white White</td></tr><tr><td>1 1/8 /12.5%</td><td>1 1/8 /12.5%</td><td>2 ¼ / 25%</td><td>2 ¼ / 25%</td><td>1 1/8 /12.5%</td><td>1 1/8 /12.5%</td></tr></table> 1 mark for correct genotypes 1 mark for corresponding phenotypes even if the genotypes are incorrect as long as they correspond 1 mark for ratios				RRGG	RWGG	RRGA	RWGA	RRAA	RWAA	Red green	Red/white green	Red variegated	Red/white variegated	Red white	Red/white White	1 1/8 /12.5%	1 1/8 /12.5%	2 ¼ / 25%	2 ¼ / 25%	1 1/8 /12.5%	1 1/8 /12.5%			3	3	3	3
RRGG	RWGG	RRGA	RWGA	RRAA	RWAA																										
Red green	Red/white green	Red variegated	Red/white variegated	Red white	Red/white White																										
1 1/8 /12.5%	1 1/8 /12.5%	2 ¼ / 25%	2 ¼ / 25%	1 1/8 /12.5%	1 1/8 /12.5%																										
	(b)			A. Seed can still germinate because of food stores/eq (1) Any four (×1) from: B. AA leads to white leaves/white leaved {plants/ phenotypes} do not grow (1) C. so no {chlorophyll/chloroplasts} (1) D. (so) no {photosynthesis/synthesis of carbohydrate} (1) E Plant dies when food stores (in seed) run out/eq (1) F. ¼ of the plants die/ white leaved plants not included in ratio/ 2: 2:1:1 ratio (, not 2:2:1:1:1:1 ratio) (1)						1 4	5																				

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					AO1	AO2	AO3	Total	Maths	Prac
	(c)			- asexual reproduction (1) - Genetically identical/ same genetic make-up/clone/same DNA (1) - Produced by mitotic cell division/ no {meiosis/gametes/ crossing over} (1)		3		3		
				Question 3 total	0	5	8	13	3	5